Assignment 5&6
Name: Digvijay Sawant
PRN: 260240125023

1. Write a script to calculate factorial of a number using while loop.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_1.sh
#!/bin/bash
read -p "Enter the number: " num
fact=1
i=$num
while [ $i -gt 1 ]; do
        fact=$((fact*i))
        i=$((i-1))
done
echo "Facorial of $num is $fact"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_1.sh
Enter the number: 5
Facorial of 5 is 120
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

2. Write a script to display first 20 even numbers using for loop.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_2.sh
#!/bin/bash
echo "First 20 even numbers:"
for ((i=1; i<=20; i++))
do
  echo $((i * 2))
done
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_2.sh
First 20 even numbers:
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

3. Write a script to print the multiplication table of a number given by the user.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_4.sh
#!/bin/bash
echo "Enter filename:"
read file
if [ -f "$file" ]; then
    lines=$(wc -l < "$file")
    words=$(wc -w < "$file")
    chars=$(wc -m < "$file")
    echo "Lines: $lines, Words: $words, Characters: $chars"
else
    echo "File does not exist."
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_4.sh
Enter filename:
file.txt
Lines: 25, Words: 231, Characters: 1290
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

4. Write a script to count lines, words, and characters in a file.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_3.sh
#!/bin/bash
echo "Enter a number:"
read n
for i in {1..10}
do
  echo "$n x $i = $((n * i))"
done
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_3.sh
Enter a number:
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

5. Write a script to display a menu (date, calendar, list files, exit) and execute the chosen option.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_5.sh
#!/bin/bash
while true; do
    echo "1. Date 2. Calendar 3. List Files 4. Exit"
    read -p "Choose an option: " choice

    case $choice in
        1) date ;;
        2) cal ;;
        3) ls ;;
        4) exit ;;
        *) echo "Invalid option" ;;
    esac
done
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_5.sh
1. Date 2. Calendar 3. List Files 4. Exit
Choose an option: 1
Monday 09 March 2026 09:39:42 PM IST
1. Date 2. Calendar 3. List Files 4. Exit
Choose an option: 4
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

(Linux Shell Programming continued)

1. Write a shell script that accepts a filename as an argument and displays whether it is a directory, a regular file, or something else.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_6.sh
#!/bin/bash
if [ -z "$1" ]; then
    echo "Please provide a filename as an argument."
    exit 1
fi

if [ -d "$1" ]; then
    echo "$1 is a Directory."
elif [ -f "$1" ]; then
    echo "$1 is a Regular File."
else
    echo "$1 is something else."
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_6.sh
Please provide a filename as an argument.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_6.sh file.txt
file.txt is a Regular File.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

2. Write a script that reads a number and prints whether it is positive, negative, or zero.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_7.sh
#!/bin/bash
read -p "Enter a number: " num
if [ $num -gt 0 ]; then
    echo "Positive"
elif [ $num -lt 0 ]; then
    echo "Negative"
else
    echo "Zero"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_7.sh
Enter a number: 5
Positive
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

3. Write a script that accepts a username and checks whether the user is currently logged in.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_8.sh
#!/bin/bash
read -p "Enter username: " user
if who | grep -w "$user" > /dev/null; then
    echo "User $user is logged in."
else
    echo "User $user is NOT logged in."
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_8.sh
Enter username: digvijay
User digvijay is logged in.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

4. Write a script that prints the first 10 natural numbers using until loop.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_9.sh
#!/bin/bash
i=1
until [ $i -gt 10 ]
do
    echo $i
    i=$((i + 1))
done
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_9.sh
1
2
3
4
5
6
7
8
9
10
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

5. Write a script that demonstrates use of functions by creating one for addition and one for subtraction.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cat script_10.sh
#!/bin/bash
add() { echo "Sum: $(($1 + $2))"; }
sub() { echo "Difference: $(($1 - $2))"; }

read -p "Enter two numbers: " a b
add $a $b
sub $a $b
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ ./script_10.sh
Enter two numbers: 10 5
Sum: 15
Difference: 5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

(Git / GitHub)

1. Install Git and configure username and email.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ sudo apt install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 299 not upgraded.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ git config --global user.name "Digvijay1508"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ git config --global user.email "digvijaysawant136@gmail.com"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$
```

2. Create a local repository, add a file, and commit changes.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5/my-repo
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ mkdir my-repo && cd my-repo
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/digvijay/Assignment/Assignment_5/my-repo/.git/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ echo "Hello Git" > file.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git add file.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git commit -m "Initial commit"
[master (root-commit) 2d66b2a] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$
```

3. Create a new branch, switch to it, make changes, and merge it with main.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5/my-repo
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ echo "New feature" >> file.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git add .
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git commit -m "Added feature"
[feature-branch 3ee014a] Added feature
 1 file changed, 1 insertion(+)
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git checkout main
error: pathspec 'main' did not match any file(s) known to git
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git merge feature-branch
Already up to date.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git branch
* feature-branch
  master
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git checkout master
Switched to branch 'master'
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git merge feature-branch
Updating 2d66b2a..3ee014a
Fast-forward
 file.txt | 1 +
 1 file changed, 1 insertion(+)
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$
```

4. Push the local repository to GitHub using command line.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5/my-repo
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git remote add origin https://github.com/digvijay1508/demo.git
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git branch -M main
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$ git push -u origin main
Username for 'https://github.com': digvijay1508
Password for 'https://digvijay1508@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 466 bytes | 466.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/digvijay1508/demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/my-repo$
```

5. Clone a repository from GitHub, edit files, and push changes back.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_5/demo
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ git clone https://github.com/digvijay1508/demo.git
Cloning into 'demo'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5$ cd demo/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/demo$ echo "Editing file" >> README.md
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/demo$ git add .
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/demo$ git commit -m "Edited README"
[main 9ac4f44] Edited README
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/demo$ git push origin main
Username for 'https://github.com': digvijay1508
Password for 'https://digvijay1508@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 300.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/digvijay1508/demo.git
   3ee014a..9ac4f44  main -> main
digvijay@digvijay-virtualbox:~/Assignment/Assignment_5/demo$
```